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Design of a Wireless Cyber-Physical System for Gas Leak Detection Using LoRa Technology

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Submitted by

Project Associates	Regd. Nos
BEERAM YESWANTH REDDY	22091A04R4
MEKAPOGU VINAY	22091A04Q9
JINKA VENKATA SAI CHARAN	22091A04Q0

Under the Esteemed Guidance of

Mr. N. Vinod Kumar

M. Tech (Ph.D.)

Assistant Professor

(ESTD-1995)

DEPARTMENT OF

ELECTRONICS AND COMMUNICATION ENGINEERING

RAJEEV GANDHI MEMORIAL COLLEGE OF ENGINEERING AND TECHNOLOGY

(AUTONOMOUS)

Affiliated to JNTUA - Anantapuramu, Approved by AICTE - New Delhi,

Accredited by NBA - New Delhi, Accredited by NAAC with A+ Grade – New Delhi

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RAJEEV GANDHI MEMORIAL COLLEGE OF ENGINEERING & TECHNOLOGY

AUTONOMOUS

(Approved by A.I.C.T.E - New Delhi, Affiliated to JNTUA - Anantapuramu,
Accredited by NBA - New Delhi, Accredited by NAAC with 'A+' Grade - New Delhi)

NANDYAL – 518501 (Dist.), AP, India

CERTIFICATE

This is to certify that the dissertation entitled **"Design of a Wireless Cyber-Physical System for Gas Leak Detection Using LoRa Technology"** is being submitted by BEERAM YESWANTH REDDY (22091A04R4),MEKAPOGU VINAY (22091A04Q9), JINKA VENKATA SAI CHARAN (22091A04Q0), under the guidance of **Mr. N. Vinod Kumar, Assistant Professor** for Project of the award of B.Tech Degree in Electronics and Communication Engineering, Rajeev Gandhi Memorial College of Engineering and Technology, Nandyal (Autonomous) (Affiliated to JNTUA Anantapuramu) is a record of bonafide work carried out by them under our guidance and supervision.

Dr. J. Sofia Priya Dharshini
M. Tech, Ph.D., MIE, MIEEE
Professor & HOD of ECE
Head of the Department

Mr. N. Vinod Kumar
M. Tech (Ph.D.)
Assistant Professor
Project Guide

Signature of the External Examiner

Date of Viva-Voce:

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Project Associates	**Regd. Nos**
BEERAM YESWANTH REDDY	22091A04R4
MEKAPOGU VINAY	22091A04Q9
JINKA VENKATA SAI CHARAN	22091A04Q0

CANDIDATE'S DECLARATION

We hereby declare that the work done in this project titled **"Design of a Wireless Cyber-Physical System for Gas Leak Detection Using LoRa Technology"** submitted towards completion of main project in IV Year II Semester of B. Tech (ECE) at the **Rajeev Gandhi Memorial College of Engineering & Technology (Autonomous), Nandyal**. It is an authentic record our original work done under the guidance of **Mr. N. Vinod Kumar, Assistant Professor**, Dept. of ECE, RGMCET, Nandyal. We have not submitted the matter embodied in this major Project for the award of any other degree in any other institution.

Project Associates	**Regd. Nos**	**Signatures**
BEERAM YESWANTH REDDY	22091A04R4	
MEKAPOGU VINAY	22091A04Q9	
JINKA VENKATA SAI CHARAN	22091A04Q0	

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ABSTRACT

Gas leaks can be extremely dangerous, especially in industrial and residential settings where early detection is crucial to prevent accidents. In this work, we present a smart and energy-efficient wireless system designed to detect gas leaks in real time. The system uses gas sensors connected to a microcontroller that continuously monitors the air for hazardous gases like methane, LPG, and carbon monoxide. If abnormal levels are detected, the system sends an alert using LoRa an advanced wireless technology known for its long-range, low-power communication. This makes it ideal for large areas or remote locations where traditional Wi-Fi or cellular networks might not be reliable. The system not only sends real-time alerts to a central monitoring unit but also keeps a record of the readings for future analysis and maintenance planning.

KEYWORDS: Wire less, Gas Sensors, Long Range Technology(LoRa).

CHAPTER-I

INTRODUCTION

1.1 INTRODUCTION

Gas leaks present serious hazards to individuals, business facilities (commercial) and both public and private structures (residential) and to many industries (agriculture) because, depending on their location and quantity, they have the potential to cause fires, explosions and/or health problems due to exposure to a toxic gas (e.g. liquefied petroleum gases (LPG), methane, carbon monoxide, etc.). The timely detection of a gas leak greatly reduces the likelihood that someone will die, sustain an injury or suffer property damage as the result of a gas leak.

Traditional gas leak detection systems have limited communications and scalability in underground areas, and rely mostly on older wire infrastructure or short-distance wireless options such as Bluetooth or Wi-Fi to communicate, limiting their usefulness in large industrial applications due to high power consumption and poor coverage area.

The growing number of applications that use Wi-Fi and other wireless connections is encouraging the development and use of IoT smart monitoring technology. These technologies are very desirable because they can gather information, monitor remotely and provide near real-time intelligent decision-making support. Of all the wireless communication technologies currently used and applicable to IoT connectivity, LoRa (long range) appears to be the only effective means of wireless communications.

1.2 Existing System

Conventional gas detectors have long been used in many industry and home locations to detect and count the hazardous gases that may be present around those facilities. A typical gas detector uses a gas sensor (i.e., the detection hardware) that is connected by some type of wired connection to a central location (usually a computer workstation) for the purpose of monitoring the sensor. Conventional gas detection systems typically have just basic functions for simply detecting gas; they have a number of limitations (for example, inflexibility, difficulty of installing, and expensive to maintain) that make them impractical for many uses.

As wireless technology has improved over the last few years, gas detection systems based on Wi-Fi, Bluetooth, and ZigBee have become available. Regardless of their popularity, these systems also have limitations with respect to distance and amount of power they consume. For example, Wi-Fi systems typically use a lot of power, while Bluetooth and ZigBee systems require the user to be relatively nearby to the device to be able to communicate.

In recent years, the introduction of IoT has created opportunities for using wireless sensor networks for environmental monitoring. LoRa (Long Range) is one of many communication technologies currently available for environmental monitoring; however, LoRa is quickly growing in use and reliability due to its capacity for long-range communication with very low power. Therefore, this technology has a great deal of promise for gas detection applications within large industrial sites.

1.3 Need for the Proposed System

The number of industrial and residential accidents arising from combustible and hazardous gases, caused by leaks, is rapidly increasing due to global industrialisation and urbanisation. Although there are multitude combustible and hazardous gases produced being used, each type of combustible and hazardous gas can create a catastrophic accident should they leak. Accidents caused by combustible or hazardous gases may result in a total loss of property as well as both environmental and life loss; therefore, it has become increasingly important to supply residential and industries with early-warning leak detection systems.

The majority of gas leak detection systems are currently based on conventional wired or short-range wireless technology so they cannot be effectively monitored over a large, distributed area. The installation of conventional systems is a complicated process that limits the flexibility of most of them to be monitored on a continuous, real-time basis. The collection of gas leak detection data from conventional systems requires that the data be stored, preventing the remote access to the information that is essential for monitoring the ambient gas concentration levels from long distances. While the use of newer, wireless technologies such as Wi-Fi and

Bluetooth allow for increased flexibility in configuring the gas leak detection system, the majority of these wireless systems exhibit similar limitations to those of conventional systems; they often experience very limited communication range or capabilities.

1.4 Problem Statement

Although there are many different types of gas detection devices available today, efficient, dependable and scalable methods of monitoring these substances remain difficult to establish. For many industrial and home users, gas leaks are not detected due to current limitations with gas monitoring methods. Traditional methods are primarily limited to wired communications systems that can be difficult and costly to install and maintain while also offering little-to-no flexibility in terms of large area coverage.

Wireless gas detectors employing Wi-Fi, Bluetooth and other wireless options have improved flexibility; however, they tend to have limited ranges and consumes large amounts of power which limits their use in long-term monitoring applications or applications requiring continuous monitoring. Furthermore, most gas detection devices currently available on the market do not permit real-time access to cloud based integration data storage and advanced level of analysis regarding the environment were the detection device is deployed.

One more major negative characteristic is the delay in generating an initial alarm and lack of usable alarm function; these delays can result in serious consequences in the event of a hazardous gas leak. The absence of a dependable, low power communication method will also contribute negatively to the overall performance of all Thus, the need exists to create an effective gas leak detector system that overcomes these obstacles by providing long range communications, using minimal power consumption along with real-time monitoring capabilities.

1.5 Objectives

design and build a wireless and cloud-based gas leak detection system using LoRa technology that can provide reliable and timely detection of hazardous gases. The overall objective of the system is to minimize the

potential risk of fire, explosion, and environmental hazardous conditions by providing reliable, early detection of gas leaks.

The project will develop a combined gas sensing, wireless communications, and cloud-based monitoring system to increase safety and operational efficiency for gas applications. The MQ-2 gas sensor will be used to continuously monitor for the presence of combustible gases, such as LPG, methane, and smoke in the surrounding air. The sensor data from the MQ-2 will be processed through the ESP32 microcontroller, which is the primary controller of the system.

The other primary goal of the project is to provide long range communication through LoRa technology. Unlike traditional wireless technologies, LoRa provides the ability to send data over long distances, with very low power consumption. This makes LoRa suitable for industrial and remote monitoring applications. The intent is to provide reliable communication between the sensor and the gateway node through efficient transmission of sensor data from the sensor to the gateway node, even in challenging environments.

The second major goal of this project will be to create a secure, 24-hour a day gas monitoring system using cloud technology. The cloud technologies allow for near real time access to sensor data from the field and provides the ability to easily create, store, share, and access historical sensor data in order to help with making decisions for future development and operation of gas sensing systems in the field.

1.6 Motivation

There is an increasing number of accidents due to gas leaks in a variety of settings. Accidents involving leaking combustible gases, such as LPG and methane, can lead to serious injury and/or death as well as property damage, usually because they were not detected soon enough or monitored adequately.

Many current gas detection systems use traditional technology, which does not always provide the necessary level of performance required by the rapidly changing safety needs of many industries. They do not have real-time monitoring capabilities, have problems with long-range

communications, and offer little or no remote access capabilities. As the number of industries and the complexity of their environments continue to increase, there is a pressing need for new, smarter, more efficient monitoring solutions that can provide continuous operation and instant notification.

The rapid growth of Internet of Things (IoT) technology has created new opportunities to develop intelligent monitoring systems. Through the use of sensors, wireless communications, and cloud-based platforms, these systems can detect hazardous conditions in real time and notify users immediately after the occurrence. Of the multiple forms of communication technologies available today, the LoRa communication technology is being relied on heavily due to the long-range communication capabilities, very low power consumption (battery-operated sensors), and suitability for large-scale monitoring applications.

1.7 Advantages of the Proposed System

A wireless and cloud solution for monitoring gas leaks through the use of LoRa technology has many benefits over traditional methods of tracking gases. One of those advantages is the range of capabilities provided by the use of a low power consumption medium that allows for long range communications (up to several kilometers away from the central unit). Therefore, this technology is well suited for large industrial locations or remote monitoring sites.

Another advantage to this gas monitoring system is its ability to continuously report on environmental conditions in real time (e.g., gas concentrations, temperature and humidity) because the IoT and Cloud platforms have integrated this continuous flow of information into the web and mobile dashboards. Therefore, the ability of this system to detect hazards will be enhanced. In addition, the ability of the system to generate alarms instantly when gas levels exceed the threshold limit will improve its reliability and safety by providing people with a warning before a catastrophic event may occur (e.g., fire and explosion) due to the leakage of gas. Furthermore, the proposed system provides a cost-effective solution by

using inexpensive components (e.g., ESP32, MQ-2 sensor, LoRa) and will allow for scalability; i.e., the addition of new sensors as necessary.

1.8 Disadvantages of the Proposed System

The proposed system has many advantages but also several disadvantages. First, the system is dependent upon the availability of internet connectivity to access data from sensors through the internet. Connection issues will halt data transmission.

Second, the temperature and humidity impact the accuracy of the readings provided by the MQ-2 Gas Sensor, but following the correct calibration procedure can provide very accurate gas detections.

Third, the proposed system will be somewhat complex to design and implement because of its use of multiple technologies (LoRa, Cloud based, and IoT) to complete a solution. Therefore, obtaining the necessary technical abilities to successfully develop and implement a solution will be needed.

Finally, LoRa has excellent range for communicating but poor data rates. Therefore, it would be better suited for applications that do not require high-speed transmission of large volumes of data (such as environmental monitoring), rather than applications that do require high-speed transmission of large volumes of data.

1.9 Applications of the Proposed System

The suggested solution has the potential to be widely utilized in applications that require gas monitoring and ensuring safety. It would be applicable to many kinds of industrial environments, including oil refineries, chemical plants and all types of manufacturing repairs, where there is the presence of hazardous gases.

In residential uses, the product could be employed as a smart safety device that can assist in identifying gas leaks in homes and avoid potential hazards. It could also be incorporated into industrial mining operations, where monitoring of dangerous toxic/harmful gases is essential for ensuring worker safety. In addition, the concept can be utilized in environments to monitor the quality of the air and determine whether or not there are any harmful gases present in the environment. Integration with cloud

applications will allow for the product to be monitored remotely, which will allow it to be applicable to smart cities and IoT driven solutions.

1.10 Organization of the Report

This report is made up of multiple parts to provide detailed information about the proposed system. The introduction section provides introductory information about the project, including background context, need for the project, problem definition, goals for the project, and reasons why this project will be pursued (Chapter 1). A literature literature review of other systems available to assist in justifying implementing this system is discussed in Chapter 2. The system design/architecture of the proposed system including the hardware and software used is discussed in Chapter 3. The methods used to implement the proposed system (Chapter 4) will consist of hardware configuration, communication methods, and implementation methodology.

Chapter 5 shows the results of the performance measurement analysis. Finally, the conclusion for this report will discuss the future scope of the project (Chapter 6).

CHAPTER-II
LITERATURE REVIEW

C. W. O'Brien, H. Salamy, and C.-H. Min, "Design of a Wireless Cyber–Physical System for Gas Leak Detection with LoRa," IEEE Internet of Things Journal, 2025. [1]

In this study, authors create a novel gas leak detection system that integrates physical and cyber components of the utility industry using wireless technology to provide high accuracy and low-cost sensors for real-time measurement. This new system consists of gas sensors coupled with microcontrollers that monitor environmental parameters consistently. The LoRa wireless communication provides long range and low power consumption by allowing remote connections to transmit sensor data over great distances. Authors also emphasize that the implementation of Low Power Wide Area Network (LoRaWAN) technologies have the potential to offer reliable communications even in very large scale scenarios. However, the primary focus of this system is on data transmission and while data visualization via cloud computing and component alerts are lacking from this device type.

H. Elazhary, A. Mahmoud, and M. Abdallah, "LoRa-based IoT System for Environmental Monitoring Applications," IEEE Access, 2023. [2]

An overview of an environment monitoring system that employs LoRa-based Internet of Things (IoT) technology.

The monitoring of environmental parameters, such as temperature, humidity, and air quality, is accomplished using an array of wireless sensors that gather this information and transmit it over long distances via the LoRa network to a primary gateway.

The gateway receives the data, processes it, and makes it available for analyses; environmental monitoring capabilities are then implemented. The advantages of using LoRa technology were proven by examining its characteristics of range and energy efficiency.Gas leak detection systems and real-time alerts have not been included within this scope of work.

J. Lin, Y. Chen, and H. Zhang, "Performance Evaluation of LoRa Communication in Wireless Sensor Networks," Sensors, 2023. [3]

This work looked at how well (or poorly) the LoRa radio technology performed when used in Wireless Sensor Networks (WSNs). Evaluating communication range, packet loss and power consumption were the three main parameters that were of interest during the study. Overall, LoRa provided reliable communications over very large distances (several kilometres) with low levels of packet loss; therefore it has real potential to be used within IoT based applications such as remote monitoring applications. However, as the study is centred on the communication performance of LoRa it does not cover how LoRa would be implemented at application level (i.e. pump station monitoring).

S. Kulkarni and R. Patel, "Wireless Gas Monitoring System Using LoRa Technology for Industrial Safety," IEEE Conference, 2022. [4]

This study proposes an industrial wireless gas monitoring system designed to improve overall industrial safety. The system incorporates gas sensors and LoRa technology to provide real-time detection and monitoring capabilities. The findings of this research demonstrate that both early gas detection and long-range data transmission are critical elements in an industrial environment. Despite achieving accurate gas detection capabilities, cloud integration and sophisticated visualizations are lacking in the proposed system design.

A. Augustin, J. Yi, T. Clausen, and W. M. Townsley, "A Study of LoRa: Long Range & Low Power Networks for the Internet of Things," Sensors, 2016. [6]

The authors will present a detailed examination and introduction of the theory and methodologies of Long Range (LoRa) Technology as they apply to Internet of Things (IoT) devices. Techniques will be identified that illustrate how LoRa technology benefits end-users through supporting long-distance, low-power consumption-based solutions; therefore, it is ideally suited for many different types of applications that require large-scale

monitoring of multiple elements simultaneously. This research serves as a basis for deployment of LoRa-based solutions across a variety of disciplines i.e. gas measurement.

A. Gupta and P. Singh, "IoT-Based Environmental Monitoring System Using LoRa Communication," IEEE Access, 2022. [5]

A new IoT-based environmental monitoring system utilizing Long Range (LoRa) communication to transmit data collected from multiple sensors such as temperature/ humidity/ air quality over significant distances has been developed. Data from the various sensors monitoring the environmental parameters are transmitted by the sensor to a data repository or server where the information can be monitored, via a central server. The low power and long-distance capabilities of the LoRa technology provide the potential for this system to be used for remote applications, making it a cost-effective solution. The results of this study indicate that the use of wireless communication provides a significant advantage to conventional wired systems regarding performance. However, the primary application of the proposed system is to monitor the environment, while gas leak detection is not included nor are any real-time alerts required for monitoring critical paths to ensure safety.

M. Ali, S. Khan, and R. Ahmad, "IoT-Based Smart Gas Leakage Detection and Monitoring System Using Cloud Integration," IEEE Access, vol. 12, pp. 34567–34578, 2024. [9]

This paper presents a smart gas leakage detection system using IoT technology combined with cloud integration. The system uses gas sensors and microcontrollers to continuously monitor environmental conditions and detect gas leakage in real time. The collected data is transmitted to the cloud platform, where it is stored and visualized for user monitoring. The study highlights the importance of real-time alerts and remote access in improving safety in industrial and residential environments. The system demonstrates improved efficiency through cloud-based monitoring and data analysis. However, the communication method used is limited to short-range

technologies and does not utilize long-range communication such as LoRa, which restricts its application in large-scale monitoring environments.

R. Sharma, P. Verma, and A. Gupta, "Design and Implementation of LoRa-Based Wireless Monitoring System for Industrial Applications," IEEE Sensors Journal, vol. 23, no. 15, pp. 16789–16798, 2023. [10]

This paper proposes a LoRa-based wireless monitoring system designed for industrial applications. The system focuses on collecting sensor data and transmitting it over long distances using LoRa communication. It demonstrates the advantages of LoRa technology in terms of low power consumption and reliable long-range communication. The study also evaluates system performance based on communication reliability and data transmission efficiency. The results show that LoRa is highly suitable for industrial monitoring systems. However, the proposed system does not include cloud-based data visualization or real-time alert mechanisms, which limits its capability for remote monitoring and intelligent analysis.

CHAPTER-III

SYSTEM DESIGN AND ARCHITECTURE

3.1 Proposed System Architecture

A novel cloud-connected "smart" system has been designed to detect gas leaks via a low-power wide-area wireless network (LoRA). The architecture of this new system allows users to access real-time gas leak alerts easily, as well as providing long-distance communication capabilities. The sensor node will measure the environment (gas concentrations, air temperature, humidity) and transmit that information to the gateway node using low-power wide-area network (LoRA) technology. The gateway node acts as an intermediary between the sensor and the cloud service for monitoring sensor data. The cloud service will receive, store, and display the data from the sensors.

3.2 Design Methodology

This structured methodology has four distinct stages: Data Acquisition, Data Processing (in an ESP32 hardware), Data Transmission, and Data Visualization.

1. Gas concentration and environmental parameters are sensed initially by the MQ-2 and DHT11 sensor devices.
2. The ESP32 device processes the data received from the sensors.
3. Processed data is then transmitted via LoRa communication over long distances.
4. At the gateway device, the data received from the LoRa devices will be sent via the MQTT protocol to the cloud. The cloud will have both InfluxDB and Grafana systems to visualize the data presented. Therefore, there will be a reliable and efficient data transfer methodology.

3.3 System Architecture Design

There are multiple layers of a multi-layer architecture for all components in the overall sensor network:

- Sensing Layer which acts to gather data from gas concentration, temperature and humidity.
- Processing Layer will perform processing of the received sensor data via the ESP32.

- Communication Layer will relay data using LoRa (long range low power) and WiFi to the Cloud for storage, viewing and alerts from the data collected.

A modular system architecture enables each unique segment to work independently yet communicate between segments to ensure seamless interconnection; therefore, a modularly designed system enables the components to be adaptable by allowing the addition of more sensor nodes to allow for continued scalability.

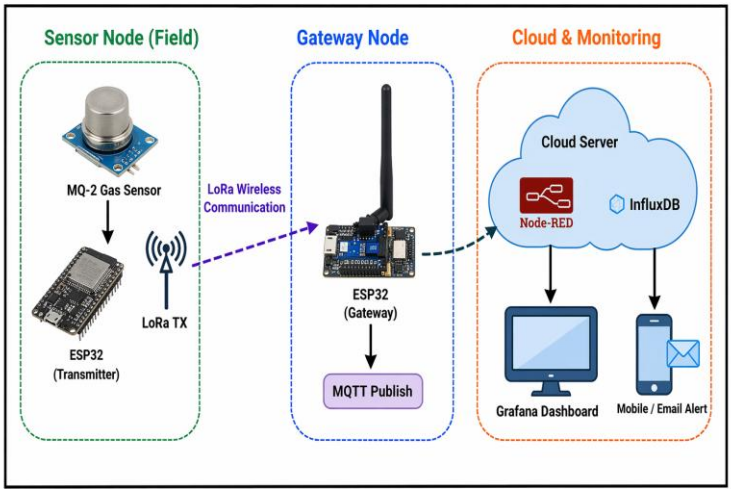


Fig 3.1: System Architecture of Proposed System

3.4 Block Diagram and Data Flow Design

This diagram depicts the whole process from data acquisition via sensing to visualizing the data into the user's hands. Data transfers across are it various steps to meet the needs of high-efficiency transmission and reliable transfer of the information.

The data acquisition progress starts with a sensor node, which acquires environmental related data and transmits it via LoRa communication. The gateway node receives the data from the sensor node and forwards the data from the sensor node to the cloud-based platform using MQTT protocol. After the data has been processed and stored in the cloud system, the data is then accessed through a dashboard for visualization.

The data flow design creates data that experiences little to no delay, has a very low amount of packet loss and allows for real-time monitoring capabilities of the sensor data.

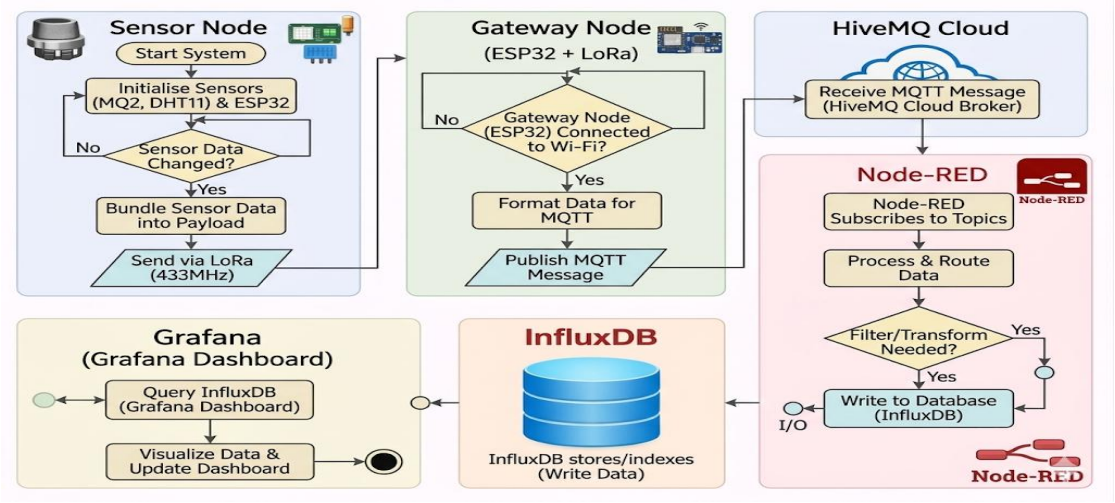


Fig 3.2: Block Diagram and Data Flow of the System

3.5 Sensor Node Design

The Data Acquisition System's Sensor Node has multiple components; an ESP32 Microcontroller, a MQ-2 Gas Sensor, a DHT11 Environmental Monitoring Sensor and a LoRa Communication Module.

The MQ-2 Gas Sensors have the ability to detect the combustibility (i.e.; LPG/Methane gases) of different gases and the DHT11 Environmental Monitoring Sensors provide important data related to the environment that may impact the ability to detect gases. The ESP32 Microcontroller was selected based on its high level of processing power and its very strong level of communication capability and the LoRa Communication Module was selected due to its ability to provide long-range communications between the ESP32 Microcontroller and the Sensor Node using the SPI Bus. The combined features of the four components of the Sensor Node provide the capability to efficiently collect and convert raw Sensor Data into usable Sensor Data which then is sent back to the Data Acquisition System.

3.5.1 Working Principle of MQ-2 Gas Sensor

The MQ-2 gas detector uses the change of resistance of metal oxide semiconductor in the presence of combustible species to measure gas concentration. This sensor measures gas concentrations using a conductive

layer of tin oxide (SnO2) to measure gas concentrations. The conductivity of the sensing element (SnO2) increases in the presence of LPG, methane, and smoke. A higher concentration of gas results in less resistance from the MQ-2; thus, the output voltage of the sensor becomes higher when gas concentration is higher.

The MQ-2 has a preheat time (approximately 30 seconds) to stabilize and perform accurately. The output of the MQ-2 sensor is an analog signal that is directly related to the concentration of gas. After the MQ-2 sensor generates an output signal, it can be read by the ESP32 microcontroller for further processing after the initial output is generated.

3.5.2 Working Principle of DHT11 Sensor

A DHT11 sensor is an electronic device that measures the temperature and relative humidity with respect to air and needs a power supply module for proper operation. This type of sensor has two major components: A capacitive humidity sensor and a thermistor (temperature sensor). It uses a capacitive humidity sensor to detect the moisture levels in the air and then provides a digital output to indicate the current temperature or relative humidity.

The DHT11 provides reliable readings for basic applications that monitor the environment, using digital signals transmitted from the sensor to an ESP32 board over a single-wire signal line interface.

3.6 Gateway Node Design

The sensor network is connected to the cloud through a gateway node, containing an ESP32 microcontroller, LoRa modem, and Wi-Fi. Data is received continuously from the sensor node into the gateway node via LoRa, and processed by the ESP32 to be sent to the cloud using MQTT. The design of the gateway node supports connectivity between long-range (LoRa) radio communications and cloud services via the Internet.

When you are processing data over the Internet, your Wi-Fi network will connect to your router, which is a device that routes packets of information across networks. Your MQTT Broker, which is another type of device that processes messages, will also be connected to your local router after both your router and your MQTT broker have been successfully

connected to each other for you to send messages to your MQTT broker from your router.

MQTT is a very lightweight protocol with very little demand on your bandwidth and therefore an excellent choice for IOT applications that require low bandwidth capabilities but require very high reliability with respect to timely transmission of data from the gateway to the cloud platform.

The gateway maintains connectivity to both the LoRa network and to your MQTT broker, which will allow for the maximum amount of time that it can maintain functionality of each system to the fullest extent possible. If your gateway loses its ability to communicate with either the Lora network or the MQTT broker due to network connectivity problems, it can automatically reconnect to the gateway or MQTT broker as soon as network connectivity is restored. This can also increase the reliability of a system because there will be a reduction in the likelihood that data will be lost. Another significant benefit of the Gateway is its ability to route data from multiple sensor nodes to the cloud. The gateway has the ability to route data from multiple sensor nodes back to the cloud using a single gateway and can operate over a large geographical area, thus providing a scalable gateway and the possibility of supporting many different types of industrial applications from one central gateway device.

3.7 Communication Design

Platforms in the communication system support LoRa (Long Range) and MQTT (Message Queuing Telemetry Transport) protocols for the provision of reliable and efficient data transmission per the requirements of a typical IoT monitoring system.

The LoRa communications protocol was chosen for its long-range characteristics and low power consumption as it operates at a low frequency in the sub-GHz frequency band and provides the capacity for the transmission of data over several kilometres.

The MQTT protocol was selected for its ability to support cloud communication via a lightweight publish-subscribe model, providing the

capability for reliable transmission between the Gate Node in the communication system and the Cloud Platform.

Sensors are the initial nodes used to communicate with the gateways. The communication link via LoRa (Long Range wireless) serves as the basis of the communication between the two devices. LoRa is a member of the LPWAN technologies and operates in the sub-GHz radio frequency (in this case at 433 MHz).CSS modulation (chirp spread spectrum) is used to maintain reliable communication and to ensure the lowest possible potential for interference. LoRa can communicate over long distances and has low power requirements regardless of obstructions (structures, buildings, industrial equipment, etc.) between the devices. Therefore, LoRa is an excellent solution to monitor gases, as they will typically have a long coverage area for deployment.

LoRa's ability to transmit data from one location to another is an enormous benefit compared to traditional wireless data transmission methods, which tend to use a great deal of power (Wi-Fi or Bluetooth). Also, battery-operated devices will benefit significantly from this method.

3.8 Cloud System Design

Cloud Computing is a fundamental component of an automatic gas leak detection system, enabling individuals to easily retrieve all relevant data from anywhere in the world. All sensor data is transmitted to the cloud where it will be saved, mapped out, and made available in real-time to users; however, the only way to see the gas detection information from the on-site system IS to view through RF, cellular, or other means.

The Cloud Architecture for this project was based on MQTT, Node-RED, InfluxDB, and Grafana. The Gateway Node sends the sensor data to the cloud utilizing MQTT Protocol, which is used mainly for IoT applications due to its low bandwidth requirements and continuous flow of data.

Data sent to the cloud is then processed through the Node-RED application, which organizes it for use and assigns any necessary condition statements (i.e. event that gas exceeds a threshold). After being processed, the data is stored in InfluxDB, a time-based database designed to store

timestamped items; therefore, it's easy to monitor and report how data has changed 식 based on timestamps.

In the Cloud, Grafana provides the visualization of the data. Grafana will provide reports and scientific statistics, and enable user interaction through development in the cloud using HTML, CSS and JavaScript.

3.9 Working Mechanism of the Proposed System

The process of operating the system involves the ability to ascertain the state of the environment (i.e. ambient conditions) on a continuous basis and to pass this data from a remote sensor(s) situated in a specified location (sensor node) to another location via the Internet of Things (sensor node to cloud). At the sensor node, the measurement of environmental states takes place (e.g. concentration of specific gases) through the use of various types of sensors (there are a number of models that can be purchased through various retailers).

The MQ-2 gas sensor surveys the concentration of gas(s) (e.g. LGP's, methane gas), while the DHT11 measures the temperature and humidity of the environment. The measurement stations (sensor nodes) are located in the environment nearby to monitor the ambient conditions changing throughout the designated area surrounding each sensor node, and will continuously transmit their associated data to the ESP32 microcontroller.

The ESP32 receives and formats the measurements, and then transmits to the gateway node (using the LoRa transmission technology) the sensor node measurements. The application of LoRa was determined to be the best choice due to its advantage of being able to transmit over great distances with low power consumption.

At the gateway node, the gateway node receives the measurements sent from its associated sensor node(s), formats them, and connects to the WLAN (wireless local area network). The gateway node will then transmit the sensor node measurements to the cloud via the MQTT protocol.

3.10 Design Considerations and Optimization

There are a lot of design factors throughout the entire design phase to provide better performance of a system.

Power efficiency reduces the amount of power used to run the LoRa communication system.

The communication method (LoRa) is long-range, providing more coverage than traditional systems. By tuning, or strengthening, the types of parameters, packet loss can be reduced, which will allow for better data reliability.

Modularity in terms of the design process supports the addition of multiple sensor nodes. Real-time monitoring is achieved through the utilization of Cloud-based connectivity. With these design considerations, the system should be able to operate effectively under all sorts of operating conditions.

The system has also been specifically designed to accommodate additional hardware as it grows over time. The amount of sensor nodes that can be used with this design will continue to increase with very little modification to the system. Using a single gateway node for all of the sensor nodes allows for one large coverage area to be created by placing multiple sensor nodes in that area.

The system also offers a way for users to maintain and store data for long periods of time. There are a number of different options available for storing large quantities of data and making it easy for users to visualize the data so they can analyze historical data and identify usage patterns. Some of these options are cloud storage solutions that leverage InfluxDB and Grafana to provide users with access to the aggregated data. Another key element of the design is the real-time alerting capabilities. When gas levels reach a potentially dangerous level, the system will immediately send out an alert to the user so that they have the opportunity to take action to keep themselves and others safe.

Overall, the various design elements of this system enable the development of a reliable and efficient system that can be used in a variety of real-world situations.

3.11 Advantages of the Proposed Design

, the solution uses a combination of the following technologies: (1) LoRa technology for long range communications; (2) high efficiency wireless

devices; (3) low-power wireless technology; and (4) cloud technologies for real-time data storage and visualizations.

By using LoRa technology, long-range transmission is possible while consuming very low power. This enhances the ability to send system-generated information to and from the remote site over extended periods (due to batteries or other remote power sources).

The information will be sent from the remote site through either direct connection via the LoRa network (in the form of radio waves), or via the internet (using other wireless means). Therefore, the communication method is highly effective, cost-effective and environmentally friendly. Using high-efficiency devices ensures device-to-device and device-to-infrastructure are fast and reliable. This ultimately provides significant cost-savings to end-users and improves the quality of service to customers.

The low-power wireless technology used ensures minimal power consumption in the overall design of the solution. Therefore, the total energy consumed by the complete solution is at a fraction of what traditional solutions consume. Using cloud-technologies ensures that the overall solution is much better designed than traditional devices due to their ease of use, low cost, scalability and the availability of a large number of users.

The combination of the internet of things will allow organisations to improve their response time to abnormal conditions (before they become operational problems), to track multiple locations from one dashboard, and to view real-time data either directly from remote locations or through the internet.

CHAPTER-IV

IMPLEMENTATION

4.1 Introduction

Understanding the process involved in the implementation of the wireless and cloud-based gas leak detection system outlined in this chapter. The implementation is mainly concerned with integrating the various sensing devices, long-range communication using LoRa technology, and cloud processing to establish an efficient method of obtaining real-time data. The resulting product will provide a system that will operate continuously, thereby providing sufficient service for detecting hazardous gases and notifying operators when these types of hazards occur.

Three main components make up the complete implementation of the gas leak detection system. These include the sensor node, the gateway node, and the cloud platform. The sensor node uses an ESP32 microcontroller for acquiring temperature, humidity, concentration of the gases present in the environment and to process that environmental information. The sensor node sends this information to the gateway node through LoRa communication. The gateway node receives the environmental data that has been transmitted from the sensor node and forwards this data to the cloud platform over Wi-Fi using the MQTT protocol.

Even though the hardware that constitutes the various components of the gas leak detection system are widely available and have been used in numerous applications, the true implementation of the system will most likely involve how well the various forms of technology for communication and as well as the cloud technologies will be able to be integrated effectively. LoRa Communication provides the ability for transmitting information over much larger distances than short-range/wireless systems by using very little power. Therefore, the system will be able to be used for large-scale monitoring applications.

4.2 Hardware Implementation

The sensor node and gateway node represent the two major components of the proposed system's hardware unit. These hardware units were designed to be as functional and simple as possible, allowing for

maximum efficiency and reliability in the acquisition and transmission of data; therefore, they were constructed using commonly available components.

The sensor node consists of a number of different devices which are used to monitor environmental conditions; these include the MQ-2 gas sensor (to detect flammable gases such as propane and methane), a DHT11 sensor (to monitor temperature and humidity), and a microcontroller (ESP32). The data from each of these devices is processed by the ESP32 and will be transmitted to the gateway node using a LoRa radio transmitter (SX1278) via an SPI protocol.

Once received by the gateway node, the data will be forwarded to a cloud computing platform. The gateway node comprises an ESP32 microcontroller and a LoRa radio transmitter all together in one device that is also connected to Wi-Fi. The LoRa module on the gateway node receives the data that was sent from the sensor node, while the ESP32 receives, processes, and sends the data to the cloud using the MQTT protocol.

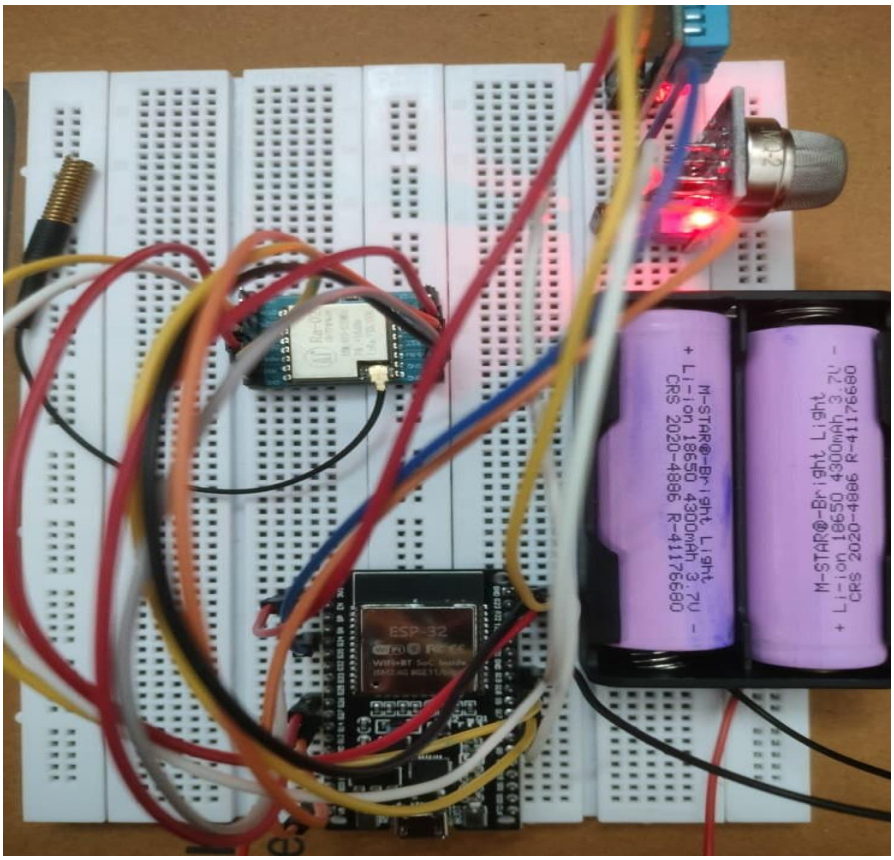


Fig 4.1: Hardware Implementation Setup

4.3 LoRa Communication Implementation

The proposed detection system for gas leaks will utilize the LoRa communication system, which utilizes wireless technology to perform long-range and low-power communications with both wired and wireless sensors. Unlike other wireless communications technologies such as Wi-Fi and Bluetooth, which have a limited range and much greater energy requirements than the LoRa communication system, the LoRa communication system allows for an extremely efficient transmission of information over very long distances. Therefore, much greater distances can be transmitted over a large distance with very little power. The proposed gas leak detection system will make use of an SX1278 LoRa module connected to the ESP32 microcontroller via a serial peripheral interface (SPI), providing reliable communication at a high speed with the LoRa device. The 433 MHz frequency of operation of the LoRa module will enhance the distance over which communication can occur as it provides superior signal propagation and ability to penetrate through physical barriers compared to other wireless technologies.

The configuration of the three (3) primary parameters must be completed in order for the performance of the proposed gas leak detection system to be optimum, as the parameters of Bandwidth (BW), Spreading Factor (SF) and Coding Rate (CR). Bandwidth (BW) determines how quickly data can be transmitted, while the Spreading Factor (SF).

Table 4.1: LoRa Module Transmission Parameters

Parameter	Value	Description
Frequency	433 MHz	Operating frequency
Bandwidth	125 kHz	Data transmission rate
Spreading Factor	SF7	Controls range & speed
Coding Rate	4/5	Error correction
Transmission Power	17 dBm	Signal strength

4.4 Gateway and MQTT Communication Implementation

The Gateway Node is known as where the two systems (LoRa sensor network and Cloud) meet together. The Gateway Node will receive LoRa Data from multiple Sensor Nodes and relay the Data to the Cloud for processing. All packets of Sensor Packet Data will be received by the ESP32 Microcontroller and processed by the ESP32 Microcontroller with the help of an SX1278 LoRa Module as the incoming hardware. After the Gateway Node receives a number of packets from the Sensors, the ESP32 Microcontroller will process the packets of Data and will send the processed packets of Data to the Cloud through the ESP32 Microcontroller.

To establish a connection to the Cloud through the processed packets of Data with the ESP32 Microcontroller, the ESP32 Microcontroller must first establish a connection to the Local Wi-Fi Network before the processed packets of Data can be sent to the Cloud. Following the establishment of a connection to the Local Wi-Fi Network, the processed packets of Data will be sent to the Cloud via an MQTT protocol (Message Queuing Telemetry Transport Protocol) as an optimal way to communicate since MQTT requires a minimal amount of power to utilize and has an extremely small footprint, thus making it ideally suited for IoT communications.

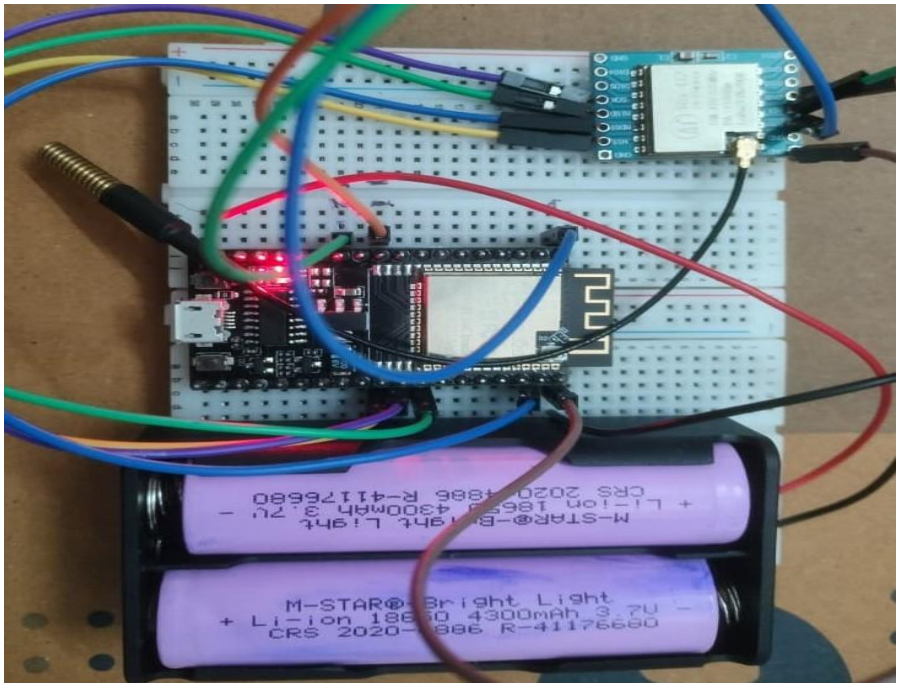


Fig 4.2: LoRa Communication between Sensor and Gateway

4.5 Cloud Implementation

The proposed system relies heavily on Cloud computing. This allows for the real-time monitoring, storage and visualisation of environmental data; easily converting the system from a basic sensor into an intelligent IoT-based monitoring platform.

The cloud application forms a core component in our proposed architecture as it provides real-time processing, storage, visual representation, and alert generation of incoming data. The system uses a variety of cloud technologies for optimal monitoring of hazardous gas concentrations with quick response times when these gases are released into the atmosphere.

4.5.1 MQTT Communication (HiveMQ Cloud)

The MQTT protocol is employed to facilitate communications between the gateway node and cloud network. As MQTT is lightweight and designed specifically for IoT devices, it provides an ideal solution for our target application. HiveMQ Cloud is the MQTT broker to publish data from an ESP32 sensor node via MQTT to the cloud. The ESP32 will publish gas concentration, temperature, and humidity readings using a pre-defined topic for each dataset. The MQTT broker will provide reliable and timely delivery of sensor data from the ESP32 to the Internet with very low latency.

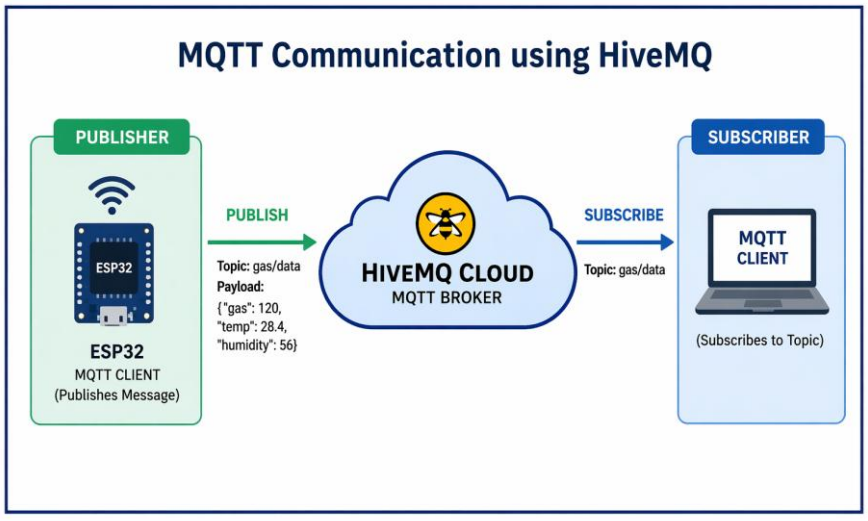


FIG 4.3: MQTT Communication using HiveMQ Cloud Broker

4.5.2 Processing Incoming Data Using Node-RED

Node-RED will be used to create a flowchart that defines how Node-RED will process the incoming data that the MQTT broker has sent to the Node-RED server via an MQTT subscription to the Node-RED MQTT broker.

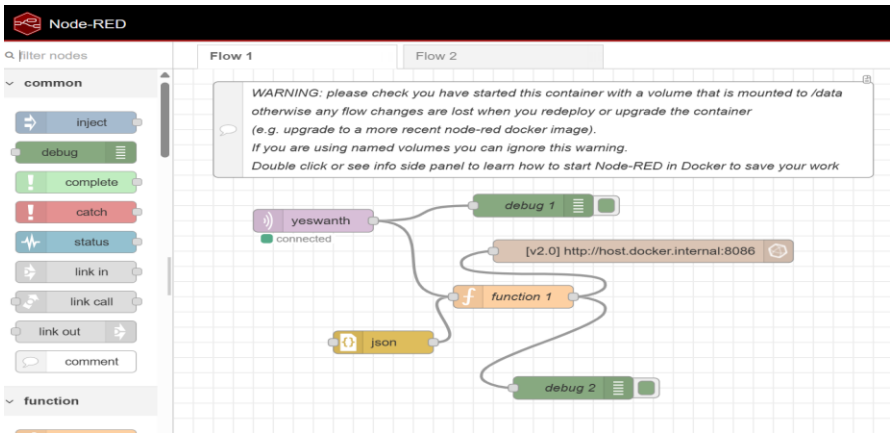


FIG 4.4: Node-Red data flow form MQTT Communication

4.5.3 Data Storage using InfluxDB

The End User (EU) has collected sensor data through their sensors. This data is kept in a time series database (InfluxDB), capable of storing timestamped data that allows for future analysis of how the data changes over time.

When the EU receives sensor updates from Node-RED, they send that data to the InfluxDB DB in a way that can be quickly retrieved and analyzed. The use of Node-RED allows them to maintain a historical record of the concentration of gas in their system and their current environmental conditions.

4.5.4 Data Visualization using Grafana

This data is presented to the user through Grafana dashboards, which use dynamic dashboards that connect to InfluxDB to visualize the captured data. The visual elements of a Grafana dashboard include graphs, gauges, and other indicators that provide users with real-time data on gas concentrations, temperatures, and humidity levels, giving them an overview of the system's operation and whether or not it is operating within acceptable standards.

4.6 Data Flow and Communication Process

The data flow for this system will facilitate the efficient and effective transfer of information collected at the sensor node level to the cloud platform. In addition to providing real-time communication of information, the communication will be accomplished through several stages and methods; using integration with LoRa communication as well as navigating the data to the cloud for storage and/or further processing.

The first stage of the communication system is the collecting of environmental data by the sensor node using an MQ-2 gas sensor and DHT11 sensor. After collecting the environmental data, the sensor will transmit this data (both gas concentration and temperature/humidity) as a structured data packet via the ESP32 microcontroller.

The data packet will be transmitted to the gateway node through LoRa transmission. LoRa communication allows the sensor to communicate to the gateway node from a long distance with a low power consumption level, providing for more reliable communication even if there is little or no line-of-sight available.

The gateway node is constantly listening for any LoRa data packets that may be received from the sensor node. When the data packet is received, it will be processed by the ESP32 and then used to create an accessible connection with the Wi-Fi network for further data transmission. Once processed, the data from the sensor node will be transmitted to the cloud through the use of the MQTT protocol.

Once the data has reached the cloud, Node-RED hosted on the cloud platform, will subscribe to the respective MQTT topics to process the incoming data. After Node-RED has processed the data from the MQTT topics, it will then store the data in the InfluxDB time series database to be utilized later for time series analysis. After the data has been stored in InfluxDB, it can be retrieved by Grafana, bringing the data back into visual dashboards for real-time visualization of the data.

4.7 System Operation

The proposed monitoring system continuously observes all environmental factors such as gas leakage. First data is collected by a

sensor node using a sensor then displayed in the cloud in real time with alerts generated accordingly.

The MQ-2 gas sensor can detect combustible gases such as LP gas and methane. At the same time as the MQ-2 sensor is obtaining readings, the DHT11 sensor will also be taking temperature and humidity measurements. Both sensors continuously monitor the environment providing output signals.

Data from both sensors will then be combined to create a unique data structure containing gas concentration in ppm, ambient temperature and humidity and is sent wirelessly by ESP32 microcontroller to the "gateway node" through LoRa communications protocol at fixed intervals.

On receiving data from the LoRa module, the ESP32 in the gateway processes the data, connects to Wi-Fi and publishes the data to the cloud over MQTT protocol.

Node-RED receives the incoming messages on the cloud using the MQTT broker and will subsequently start processing the incoming data. Node-RED controls how the data travels throughout the system.

4.8 Software Implementation

4.8.1 Sensor Node Program

Using the MQ-2 and DHT11 sensors, the sensor node program collects data regarding gas concentrations, temperature, and humidity. The data collected from these sensors will be processed by an ESP32 and will be sent from the sensor node to a gateway node via LoRa communication every predetermined number of seconds.

Code:

```
#include <SPI.h>
#include <LoRa.h>
#include <DHT.h>
#define SS 5
#define RST 14
#define DIO0 26
#define MQ2_PIN 34
#define DHTPIN 4
```

```
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
unsigned long previousMillis = 0;
const long interval = 30000;   // 30 seconds
void setup() {
  Serial.begin(115200);
  dht.begin();
  LoRa.setPins(SS, RST, DIO0);
  if (!LoRa.begin(433E6)) {
    Serial.println("LoRa init failed!");
    while (1);
  }
  Serial.println("Gas + Temp + Humidity Node Ready");
}
void loop() {
  unsigned long currentMillis = millis();
  if (currentMillis - previousMillis >= interval) {
    previousMillis = currentMillis;
    int gasValue = analogRead(MQ2_PIN);
    float temperature = dht.readTemperature();
    float humidity = dht.readHumidity();
    if (isnan(temperature) || isnan(humidity)) {
      Serial.println("Failed to read from DHT sensor!");
      return;
    }
    // Create JSON payload
    String jsonPayload = "{";
    jsonPayload += "\"gas\":" + String(gasValue) + ",";
    jsonPayload += "\"temperature\":" + String(temperature, 2) + ",";
    jsonPayload += "\"humidity\":" + String(humidity, 2);
    jsonPayload += "}";
    LoRa.beginPacket();
    LoRa.print(jsonPayload);
```

```
    LoRa.endPacket();
    Serial.println("Sent JSON: " + jsonPayload);
  }
}
```

During Setup Phase the Program initializes LoRa Modul and Sensors The ESP32 Reads Sensor Values At Intervals Using Time Based Approach And Format the Sensor Values into A Packets; Pack Billions of Sensor Values as One Pack Sent Using LoRa Communication to gateway Node via Wireless Method.

4.8.2 Gateway Node Program

The gateway node receives data from the sensor node through LoRa communication and forwards it to the cloud platform using MQTT protocol over Wi-Fi.

Code:

```
#include <SPI.h>
#include <LoRa.h>
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <PubSubClient.h>
// LoRa Pins
#define SS 5
#define RST 14
#define DIO0 26
// WiFi Credentials
const char* ssid = "DSVVY";
const char* password = "DSVVY@12";
// HiveMQ Cloud Details
const char* mqtt_server =
"2a198f5adfe84e7787662121d55c6d77.s1.eu.hivemq.cloud";
const int mqtt_port = 8883;
const char* mqtt_user = "gas_user1";
const char* mqtt_password = "Gasuser123";
WiFiClientSecure espClient;
```

```
PubSubClient client(espClient);
const char* topic = "yeswanth";
// WIFI SETUP
void setup_wifi() {
  Serial.println("Connecting to WiFi...");
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nWiFi Connected!");
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());
}
// MQTT RECONNECT
void reconnect() {
  while (!client.connected()) {
    Serial.println("Connecting to HiveMQ Cloud...");

    if (client.connect("ESP32_Gateway", mqtt_user, mqtt_password)) {
      Serial.println("Connected to HiveMQ!");
    } else {
      Serial.print("Failed, rc=");
      Serial.println(client.state());
      delay(2000);
    }
  }
}
// SETUP
void setup() {
  Serial.begin(115200);
  LoRa.setPins(SS, RST, DIO0);
  if (!LoRa.begin(433E6)) {
```

```
    Serial.println("LoRa init failed!");
    while (1);
  }
  Serial.println("LoRa Initialized Successfully");
  setup_wifi();
  espClient.setInsecure();
  client.setServer(mqtt_server, mqtt_port);
  Serial.println("Gateway Ready");
}
//  LOOP
void loop() {
  if (!client.connected()) {
    reconnect();
  }
  client.loop();
  int packetSize = LoRa.parsePacket();
  if (packetSize) {
    String received = "";
    while (LoRa.available()) {
      received += (char)LoRa.read();
    }
    received.trim();
    Serial.println("LoRa Received: " + received);
    bool ok = client.publish(topic, received.c_str());
    Serial.println("Publishing JSON: " + received);
    if (ok) {
      Serial.println("Published to HiveMQ Cloud");
    } else {
      Serial.println("Publish failed");
    }
  }
}
```

The gateway continuously listens for incoming LoRa packets. Upon receiving data, it establishes a connection with the Wi-Fi network and publishes the data to the MQTT broker for cloud-based processing and visualization.

4.8.3 Node-RED Processing Code

Code:

```
let data = msg.payload;
msg.payload = {
    gas: Number(data.gas),
    temperature: Number(data.temperature),
    humidity: Number(data.humidity)
};
msg.measurement = "environment";
node.warn(msg.payload);
return msg;
```

The Node-RED function processes incoming MQTT data and converts it into a structured numerical format. The processed data is then forwarded to InfluxDB for storage and Grafana for visualization.

CHAPTER-V

RESULTS AND ANALYSIS

5.1 Introduction

In this chapter we present results from our implementation of the proposed wireless gas leak detection system utilizing LoRa technology and cloud integration. We assess how well our system performs by evaluating three distinct categories of metrics: Communication Reliability; Data Transfer Efficiency; and Real Time Monitoring Capability. We use the experimental data collected through our testing and provide graphical representations of those results via our Cloud based dashboard.

5.2 LoRa Communication Performance

A method of measuring how well the LoRa communications system operates is through looking at how many packets can be successfully sent and received with the sensor and gateway. Indoor testing was completed to establish the reliability of data transmission.

Table 5.1: LoRa Packet Transmission Performance

Test Number	Packets Sent	Packets Received	Packet Loss	Success Rate (%)
Test 1	10	10	0	100
Test 2	20	19	1	95
Test 3	30	29	1	96.7
Test 4	40	39	1	97.5

The result showed that the transaction was successful more than 95% of the time, meaning that this system had a high success rate in transferring data from one point to another. Packet loss is small enough that it shows there is an established way to ensure reliable communication between the sensor node and the gateway node. The results also prove how well LoRa technology works for providing stable long-range communications with low power requirements.

5.3 Real-Time Monitoring Results

The system monitors environmental conditions in real-time, using sensors to report environmental data such as gas concentration, temperature, and humidity directly to the cloud. Sensor readings are available through Grafana dashboards.



Fig 5.1: Real-Time Monitoring Dashboard using Grafana

In graph format, enabling users to continuously monitor environmental conditions via real-time visualization. The dynamic update of the sensor readings aide in early identification of abnormal gasses by updating dynamically; thus enhancing usability and effectiveness of the system.

5.4 Data Storage and Visualization

InfluxDB is used by the system to save sensor data in a time-series format and Grafana helps visualize this data. Each data point is captured with a timestamp allowing the tracking of the evolution of parameters over time.



Fig 5.2: Time-Series Data Visualization using InfluxDB

The evolution of gas concentration, temperature and humidity can be seen through the data stored. This data will be instrumental in identifying trends as well as identifying deviations from normal operating conditions in the environment. The cloud-based storage of this data allows easy access to historical data for future analysis.

5.5 Remote Monitoring

The suggested system includes web and mobile dashboard-based monitoring of environmental conditions, allowing for real-time data access from anywhere that has internet coverage.

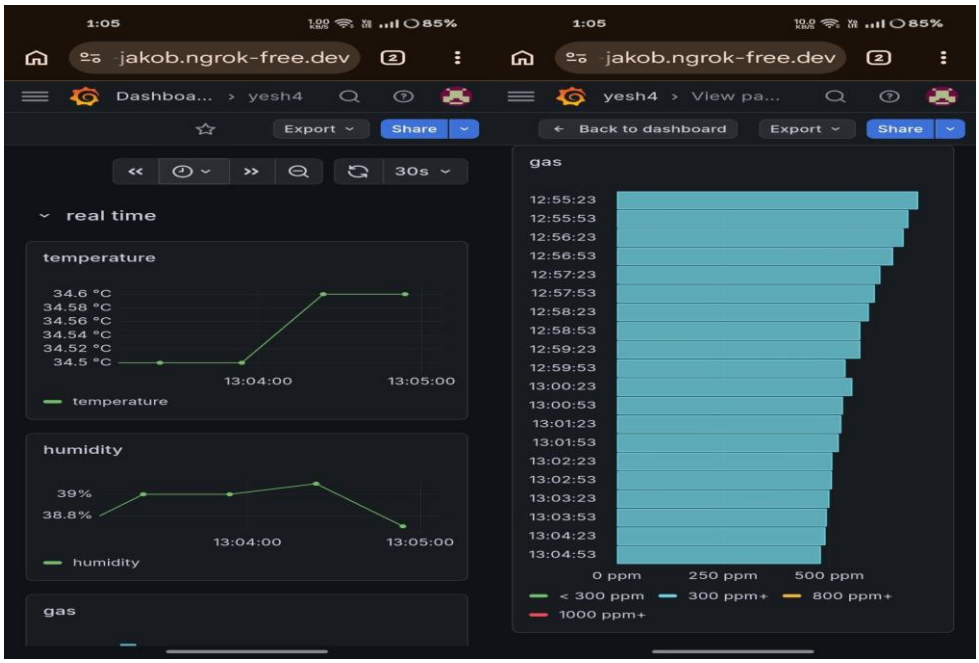


Fig 5.3: Remote Monitoring through Mobile Dashboard

The mobile dashboard improves accessibility to users due entirely to its intuitive user interface, allowing users to articulate responses quickly in the event of an adverse event.

5.6 Alert Notification System

In terms of safety issues, an alarm notification system is being developed to give alarms quickly if hazardous gas concentrations are at a threatening level or exceed a specified threshold. Any time there is a gas level above that threshold, an alarm will be sent out to the user.

The alarms will be sent out to users through email through a Node-RED process hosted in the cloud and will become activated based on the sensor(s) crossing the predetermined threshold for the level of gas.

The alarm notification sent through email to the user will have the actual level of gas detected, along with a warning of the current level of gas to warn the user so that they can take appropriate action in a timely manner. This feature allows for faster response times, thereby reducing the risk of a gas leak accident.

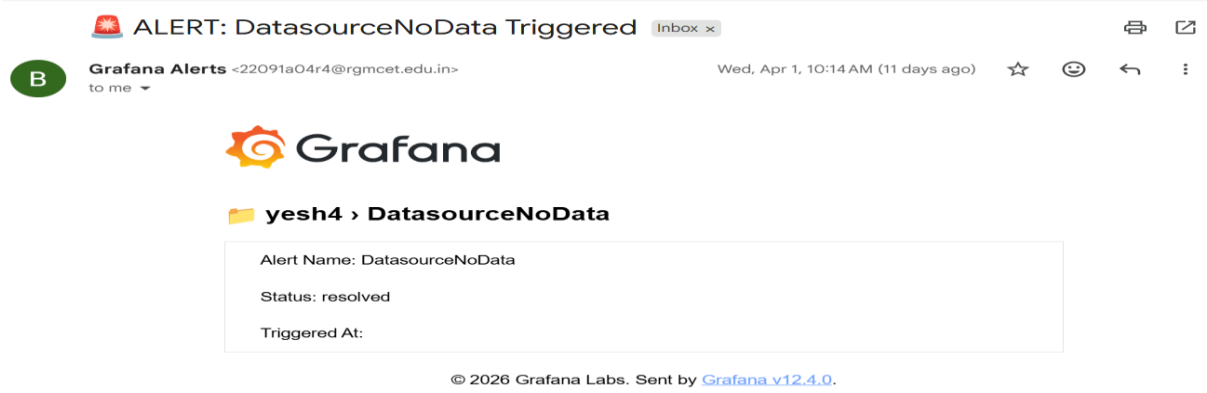


Fig 5.4: Email Alert Notification for Gas Leakage Detection

The alarm notification system allows for notifications to be sent out immediately once abnormal or out-of-normal-range levels of gas are detected, which leads to an increase in the reliability and efficiency of the overall alarm notification system. The ability to send out alarms from the cloud gives users real-time access to their accounts regardless of whether they are on site monitoring the detection devices or not.

5.7 Overall System Performance

System performance will be assessed in terms of its ability to communicate consistently, its ability to monitor data in real-time, and the level of cloud integration. All of this means the system can run continuously with very little packet loss (no more than 1%) and operate efficiently with low latency between nodes and terminals.

The use of LoRa for wireless communications combined with cloud computing provides the ability to communicate over long distances and access the most recent data in real-time. The system works well indoors and provides a timely warning whenever there is an increase of toxic gas to unsafe levels.

CHAPTER VI
CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The use of LoRa technology, a low-cost, low-power radio technology, allows long-range wireless transmissions for measuring hazardous gas levels and monitoring environmental variables (i.e., Temperature and Humidity). The wireless and cloud-based gas leak monitoring system has been designed, implemented, and operated. It successfully detects hazardous gases and continuously monitors the environment. Using the LoRa technology from the sensor node to the gateway node, the system provides long-range wireless communication for measuring hazardous gas levels and monitoring environmental variables with very low power consumption.

The successful operation of the system has demonstrated that it is feasible for monitoring large areas with a high level of reliability, producing very little packet loss.

The integration of cloud technologies has allowed for the storage, visualization, and monitoring of real-time data, with remote access to environmental data via web-based and mobile-based dashboards, enabling users to access environmental data from virtually anywhere; thus, it increases system access and usability.

The email alert notification system that was implemented provides enhanced safety for the monitoring system by giving immediate notifications to the user when an increased level of gas is detected before an accident could occur, allowing for a rapid response.

The proposed monitoring system can be implemented for several different applications. The proposed monitoring system is applicable for continuous monitoring for gas leaks in industrial settings, smart homes, and gas storage facilities. The combination of long-range wireless communications, cloud-based monitoring, and email alert notification systems makes the proposed gas leak detection system an effective and reliable safety solution.

6.2 Future Scope

Although the current system is performing well, there is room for additional improvements to increase its effectiveness.

The application can further improve its effectiveness by integrating other types of sensors to detect additional gases and other environmental parameters, which would allow for the usage of the application in different types of environments. Additionally, adding machine learning functionality to the sensors to enable sensor data analysis and the identification of gas leak patterns would enable predictive analytics for potential future gas leaks and thus allow for proactive preventative measures. Another potential approach to enhancing the user experience with the system would be through the addition of a dedicated mobile application that would provide real-time alert notifications and provide the user with the ability to control the system. Moreover, the system can also be utilized in larger industrial sites or outdoors and cover more distance and different environmental conditions. Possible ways in which to improve the system in the future include increasing the effectiveness of the notification system with alternate notification methods (such as SMS and push mobile technology), thereby improving the emergency response time. The potential use of advanced energy management techniques could also be implemented that decrease the energy usage of the system and make it more practical as a long-term solution.

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1 message

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Rajeev Gandhi Memorial College of Engineering and Technology, Nandyal
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